

BRIEF REPORT

Cardiac Arrest and Outcome in Patients With Cardiogenic Shock Complicating Acute Myocardial Infarction

A Pooled Analysis of Randomized Controlled Trials

Anne Freund, MD,^{a,*} Annika Rodewald, BSc,^{a,*} Ibrahim Akin, MD,^b Georg Fuernau, MD,^c Tienush Rassaf, MD,^d Peter Nordbeck, MD,^e Hans-Josef Feistritz, MD, PhD,^a Janine Pöss, MD,^a Marko Noc, MD,^f Ingo Eitel, MD,^{g,h} Carsten Skurk, MD,ⁱ Lars Maier, MD,^j Norman Mangner, MD,^k Taoufik Ouarrak, PhD,^l Steffen Schneider, PhD,^l Steffen Desch, MD,^a Uwe Zeymer, MD,^{b,l,m,†} Holger Thiele, MD^{a,h,†}

Cardiac arrest (CA) occurs in 28% to 77% of patients included in large randomized controlled trials (RCTs) in infarct-related cardiogenic shock (AMI-CS).¹⁻³ The DanGer-Shock trial, the first RCT showing superiority of a mechanical circulatory support device in AMI-CS, excluded patients who remained comatose after out-of-

hospital CA,⁴ renewing the discussion regarding the inclusion of CA patients in future AMI-CS RCTs. Current evidence on the prognostic impact of CA in AMI-CS remains inconsistent.^{5,6}

Therefore, the current analysis aims to compare presentation and short- and mid-term outcomes of patients with and without CA in large RCTs using a pooled data set from the ECLS-SHOCK (Extracorporeal Life Support in Infarct-Related Cardiogenic Shock), CULPRIT-SHOCK (PCI Strategies in Patients with Acute Myocardial Infarction and Cardiogenic Shock), and IABP-SHOCK (Intraaortic balloon support for myocardial infarction with cardiogenic shock) II trials.^{3,7,8}

Design and results of the trials have been published previously.^{3,7,8} In brief, patients with AMI-CS (ST-segment elevation myocardial infarction [STEMI] or non-STEMI) and planned early

What is the clinical question being addressed?

What is the impact of cardiac arrest in randomized controlled trials of AMI-CS?

What is the main finding?

Cardiac arrest patients enrolled in 3 large AMI-CS trials were younger with fewer comorbidities but longer ICU stays, while 12-month mortality was comparable to patients without cardiac arrest.

From the ^aHeart Center Leipzig at Leipzig University, Department of Internal Medicine/Cardiology and Leipzig Heart Science, Leipzig, Germany; ^bFirst Department of Medicine, University Medical Centre Mannheim, Faculty of Medicine Mannheim, University of Heidelberg, Mannheim, Germany; ^cClinic for Internal Medicine II (Cardiology, Angiology, Diabetology, Intensive Care Medicine), Städtisches Klinikum Dessau, Dessau-Rosslau; ^dDepartment of Cardiology and Vascular Medicine, West German Heart and Vascular Center, University Duisburg-Essen, Essen, Germany; ^eDepartment of Internal Medicine I, University Hospital Würzburg, Würzburg, Germany; ^fUniversity Medical Center Ljubljana, Ljubljana, Slovenia; ^gUniversity Heart Center Lübeck, Lübeck, Germany; ^hGerman Center for Cardiovascular Research (DZHK), Partner Site Hamburg/Kiel/Lübeck/Greifswald, Lübeck, Germany; ⁱUniversity Clinic Charité, Campus Benjamin Franklin, Berlin, Germany; ^jDepartment for Internal Medicine II, University Hospital Regensburg, Regensburg, Germany; ^kDepartment for Internal Medicine and Cardiology, Heart Center Dresden, Faculty of Medicine and University Hospital Carl Gustav Carus, TU Dresden University of Technology, Dresden, Germany; ^lStiftung Institut für Herzinfarktforschung, Ludwigshafen, Germany; and the ^mDepartment of Cardiology and Angiology, University Heart Center Freiburg-Bad Krozingen, Faculty of Medicine, University of Freiburg, Freiburg, Germany. *These authors contributed equally to this work as first authors. †These authors contributed equally to this work senior authors.

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revascularization were enrolled. Patients with cardiopulmonary resuscitation (CPR) >30 minutes (IABP-SHOCK II and CULPRIT-SHOCK) or >45 minutes (ECLS-SHOCK) were excluded. Full informed consent was obtained from all patients using a stepwise process. Each trial was approved by the responsible ethics committees and was performed under consideration of the principles of the Declaration of Helsinki.

For the present analysis, patients from the pooled data set were stratified according to the presence of CA defined as the need for CPR <24 hours before randomization. After excluding 3 patients with missing CPR status, 1,699 patients were analyzed.

The primary outcome was 12-month all-cause mortality with additional landmark analyses of mortality from days 0 to 30, 30 days to 6 months, and 6 to 12 months. Cumulative incidence functions comparing patients with and without CA and HRs and their respective 95% CIs were computed. HRs were further adjusted using a Cox regression model to account for potential differences in baseline and procedural characteristics. The following variables were included: age, female sex, diabetes mellitus, prior stroke, STEMI, levels of laboratory values on admission (creatinine, hemoglobin, glucose), lactate before percutaneous coronary intervention, severity of coronary artery disease, and TIMI (Thrombolysis In Myocardial Infarction) flow after percutaneous coronary intervention. To account for missing variables, mean imputations were performed for lactate and glucose levels.

Statistical analyses were performed using SAS version 9.4 (Statistical Analysis Software).

A total of 959 patients (56.3%) had CA and 740 patients (43.5%) had no CA. Patients with CA were younger (age 64 years; Q1-Q3: 56-74 years vs 71 years; Q1-Q3: 62-78 years), more often men (78.8% vs 69.9%), and had lower rates of comorbidities, prior coronary revascularization, and triple vessel disease.

The primary outcome of all-cause 12-month death occurred in 519 of 956 patients (54.3%) with CA and in 382 of 737 patients (51.8%) without CA (unadjusted HR: 1.05; 95% CI: 0.92-1.20; adjusted HR: 1.11; 95% CI: 0.95-1.29) (**Figure 1A**). Overall, 47.0% in the CA group and 43.0% in the no-CA group died between 0 and 30 days, additional 5.5% and 6.5% between 30 days and 6 months, and 1.7% and 2.3% between 6 and 12 months, respectively (**Figure 1B**). Sensitivity analyses including only patients with STEMI or non-STEMI showed comparable adjusted results. In addition, no differences between patients with and without CA were observed within each of the 3 individual trials. In both groups, cardiovascular events,

in particular refractory CS, were the most common causes of death. Fatal brain injuries occurred in 22% of deceased patients in the CA group and in 10% in the no-CA group.

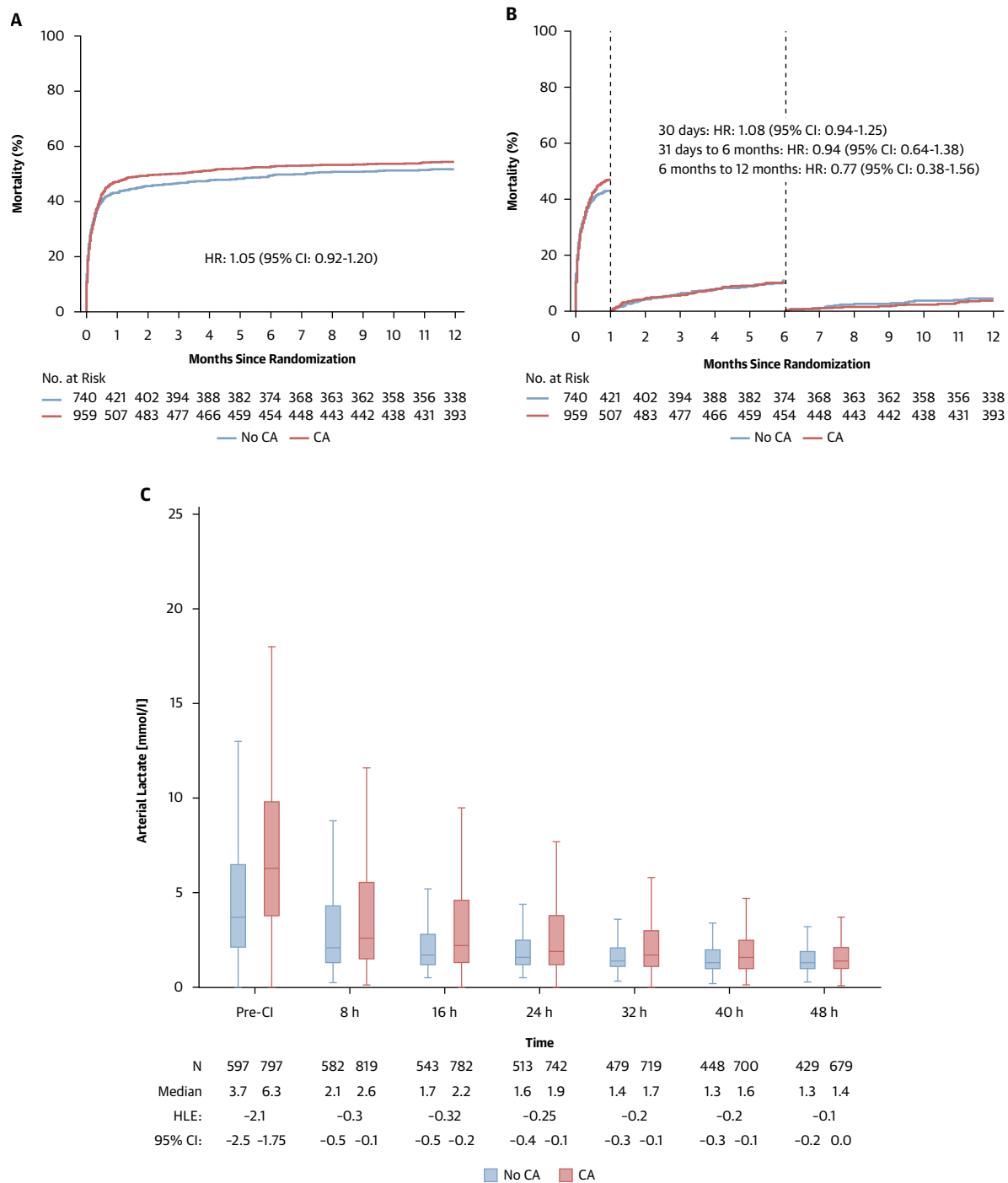
Patients with CA had a longer duration of mechanical ventilation (5 days; Q1-Q3: 2-10 days vs 3 days; Q1-Q3: 1-7 days), catecholamine therapy (4 days; Q1-Q3: 2-7 days vs 2 days; Q1-Q3: 1-5 days), and overall intensive care unit treatment (8 days; Q1-Q3: 4-14 days vs 5 days; Q1-Q3: 2-10 days). Serial lactate measurements showed higher lactate levels in the CA group throughout the first 40 hours after randomization (**Figure 1C**). Initial lactate levels were <2 mmol/L in 184 (13.2%) of patients with a higher proportion in the no CA group, but without significant differences in all-cause mortality between groups.

This secondary analysis of 3 of the largest RCTs in AMI-CS showed that more than one-half of the patients experienced CA prior randomization but did not show differences in short- and mid-term mortality compared with those without.

The proportion of patients with CA was higher compared with large registries.^{5,9} This is most likely driven by the high CA proportion in ECLS-SHOCK, which had a lactate threshold of 3 mmol/L. The observed differences in baseline characteristics between groups mirror findings from observational studies.^{5,10} Lower age and fewer comorbidities in the CA group may explain similar mortality. In contrast, patients with preexisting heart disease and triple vessel disease may be more likely to develop severe AMI-induced left ventricular dysfunction leading to systemic hypoperfusion without CA.

Overall, mortality was lower when compared with a large Danish nationwide AMI-CS.⁵ This is most likely attributable to the more selected RCT population (exclusion of patients without planned revascularization, long CPR duration, shock duration >12 hours and severe life-limiting comorbidities). Despite the limited neurologic prognostication before randomization, cardiovascular events were the most common cause of death. Associations between initial lactate and mortality in patients with and without CA have been reported separately elsewhere.¹¹

Limitations of the present analysis include the selection bias inherent in the respective trial criteria, a heterogeneous cohort with varying shock severities that cannot be determined, and the lack of details on the circumstances of CA. This also implies considerable heterogeneity within the CA group, ranging from brief CPR episodes to prolonged resuscitation and unknown no-flow times associated with high risk of hypoxic brain injury. Future trials must therefore

FIGURE 1 All-Cause Mortality and Serial Arterial Lactate Measurements in Patients With and Without Cardiac Arrest

Cumulative incidence curves for (A) the primary endpoint 12-month all-cause mortality and (B) within the landmark time points of 30 days, 6 months, and 12 months stratified by the presence of cardiac arrest (CA). Unadjusted HRs and 95% CI are displayed for (A) 12 months and (B) the respective time intervals. Results demonstrate no differences in all-cause mortality between patients with and without CA at any time interval. (C) Box plot of median arterial lactate levels prior percutaneous coronary intervention (PCI) and every 8 hours during the first 48 hours after PCI in patients with and without CA. Hodges-Lehmann Estimation (HLE) for location shift were used to calculate differences in medians and normal approximation at each time point. Patients with CA displayed significantly higher median lactate levels prior and throughout the first 40 hours after PCI.

focus on the detailed and consistent characterization of CA patients using key Utstein variables.

According to the available evidence, CA should not, by itself, mandate exclusion from future AMI-CS trials. Instead, it will require future research, particularly with regard to early neurological prognostication, to find the balance between sensible patient selection and the applicability of new treatment options for a representative AMI-CS collective. Otherwise, an important patient cohort in AMI-CS risks being left without evidence-based therapeutic recommendations.

DATA AVAILABILITY STATEMENT The data sets analyzed during the current study are not publicly available caused by privacy restrictions.

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ADDRESS FOR CORRESPONDENCE: Dr Holger Thiele, Heart Centre Leipzig at Leipzig University, Department of Internal Medicine/Cardiology, Strümpellstr. 39, 04289 Leipzig, Germany. E-mail: holger.thiele@medizin.uni-leipzig.de.

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